CASE STUDY REPORT

Battery Energy Storage System (BESS) – Case Study No 62

Subject: Business Studies – Semester IV

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GitHub Link: https://github.com/RaginiSingh2024/Battery-Energy-Storage-Business-CaseStudy

Excel File Link:
https://1drv.ms/x/c/843cce1f097d9ae9/IQBaAzoiLZGkRrGuZi2ZBR8xATFIxDoRYDgKXln_lYYVPDE?e=s1en2P

1. Executive Summary:

Power Store Solutions plans to implement a Battery Energy Storage System (BESS) in a small city facing power fluctuations and high peak-time electricity costs. The project requires an investment of ₹1 crore and aims to reduce electricity expenses, improve grid reliability, and promote renewable energy utilization. Financial analysis shows that the project can recover its investment within approximately 4–5 years and generate long-term benefits.

2. Introduction:

Power Store Solutions is planning to install a Battery Energy Storage System to manage electricity demand efficiently. The city currently experiences frequent voltage fluctuations and high electricity costs during peak demand hours. By storing excess electricity during low demand periods and supplying it during peak hours, the company can stabilize the grid and reduce operational expenses.

3. Problem Statement:

The city faces several energy challenges:

- High peak-time electricity cost
- Frequent voltage fluctuations
- Lack of energy storage for renewable sources
- High maintenance cost
- Unstable power supply

A reliable energy storage solution is therefore required.

4. Objectives:

The objectives of this project are:

- Reduce peak-time electricity cost
- Improve grid stability
- Optimize renewable energy usage
- Reduce maintenance expenses
- Achieve long-term financial sustainability

5. Financial Analysis:

Assumptions:

Annual peak electricity cost = ₹40,00,000
Maintenance cost = ₹10,00,000

Savings Calculation:

50% reduction in peak cost = ₹20,00,000
40% reduction in maintenance cost = ₹4,00,000

Total annual savings = ₹24,00,000

Investment = ₹1,00,00,000

Payback Period:

1,00,00,000 ÷ 24,00,000 ≈ 4.1 years

This means the investment will be recovered in approximately 4–5 years.

6. Technical Analysis

The Battery Energy Storage System stores excess electricity during low demand hours and releases it during peak demand hours.

Key components include:

- Battery Cells
- Battery Management System (BMS)
- Power Conversion System (PCS)
- Monitoring Software

The system monitors parameters such as battery charge level, temperature, voltage, and electricity demand to ensure efficient operation.

7. Data Simulation

To evaluate the system performance, electricity demand can be simulated using Excel. Hourly demand data is generated and peak hours are identified. During peak demand periods the battery discharges stored energy, reducing dependency on the main grid.

8. Business Impact

Benefits of implementing BESS include:

- Improved grid reliability
- Reduced electricity cost
- Better renewable energy utilization
- Lower carbon emissions
- Improved energy security for the community

9. Risk Analysis

Possible challenges include:

- High initial investment
- Battery degradation over time
- Technology maintenance cost

However, these risks can be managed with proper planning and maintenance.

Figure 1: Cost vs Savings Analysis

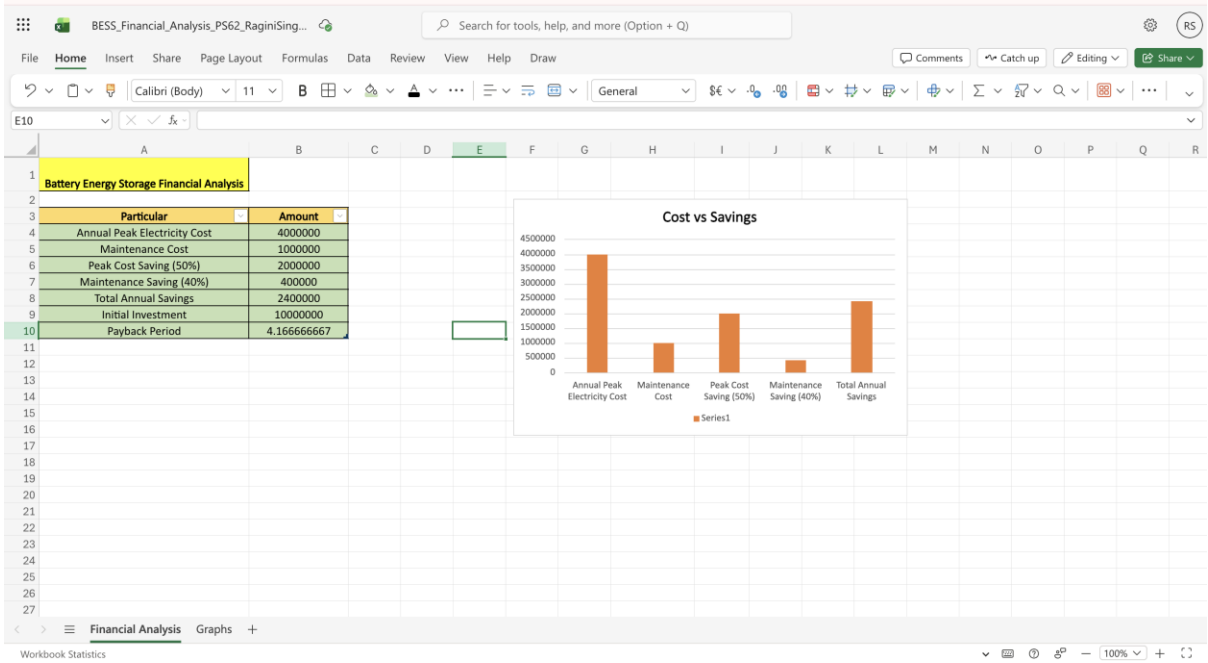
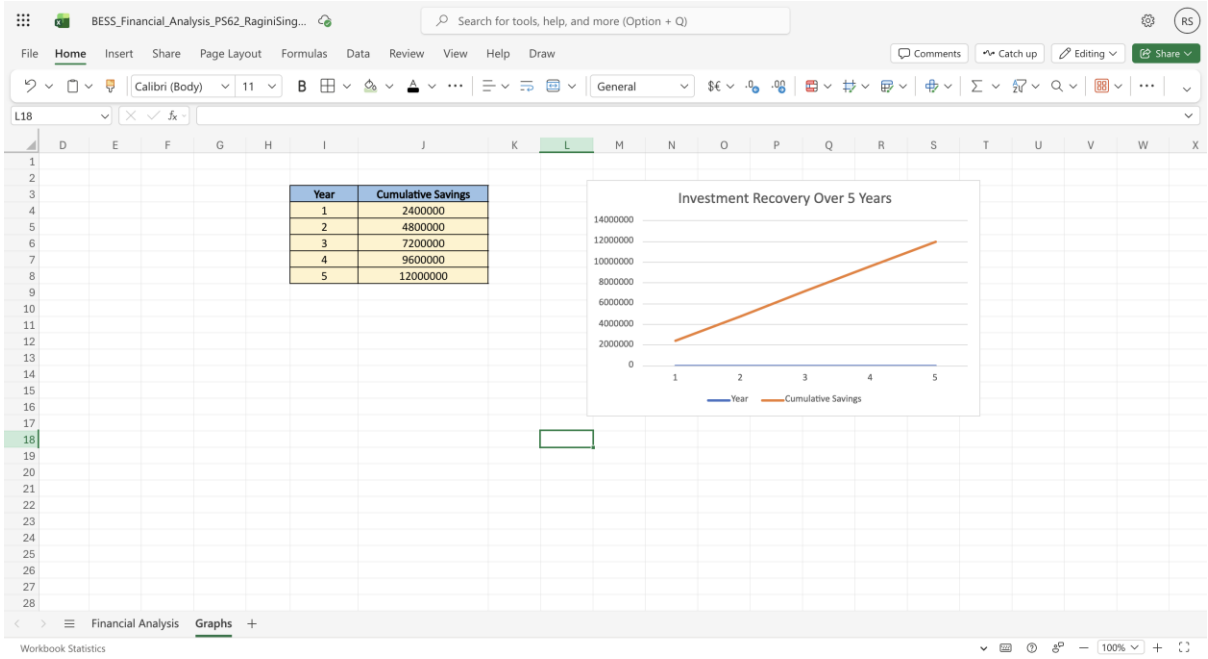


Figure 2: Investment Recovery Over 5 Years



10. Recommendation

Based on financial and technical analysis, Power Store Solutions should proceed with the BESS project because it offers long-term cost savings, improved grid stability, and sustainable energy management.

11. Conclusion

The Battery Energy Storage System is a strategic solution to reduce peak-time electricity costs and improve energy reliability. With a payback period of approximately 4–5 years and significant environmental benefits, the project is financially and operationally viable.

THANKYOU!!